Encryption System Explained

Symmetric vs Asymmetric Encryption, and How HTTPS Uses Both

Muhammad Ali Khan | Roll No: MTEC CS26409

Task Title: Encryption Explained (Sym vs Asym) | Deliverable: Explainer with a Diagram

1. Introduction

Encryption is the process of converting readable data (plaintext) into an unreadable form (ciphertext) so that only authorized parties can access the original information. There are two core approaches used in modern security systems: symmetric encryption and asymmetric encryption. Real-world systems, most notably HTTPS, combine both to get the benefits of each.

2. Symmetric Encryption

Symmetric encryption uses a single shared secret key to both encrypt and decrypt data. The same key must be known to both the sender and the receiver before communication begins.

How it works

- The sender encrypts plaintext using a secret key, producing ciphertext.
- The ciphertext is sent to the receiver.
- The receiver decrypts the ciphertext using the exact same key.

Common algorithms

- AES (Advanced Encryption Standard) - industry standard, used almost everywhere today.
- DES / 3DES - older standards, now considered weak or legacy.

Strengths and weaknesses

- Strength: Very fast and efficient, ideal for encrypting large amounts of data.
- Weakness: Key distribution is difficult - both parties need the same key, and sharing it securely over an open network is a challenge.

3. Asymmetric Encryption

Asymmetric encryption (also called public-key encryption) uses a mathematically linked pair of keys: a public key and a private key. Data encrypted with the public key can only be decrypted with the corresponding private key, and vice versa.

How it works

- The public key is shared openly with anyone.
- The private key is kept secret and never shared.
- Anyone can encrypt a message using the recipient's public key, but only the recipient (holding the private key) can decrypt it.

Common algorithms

- RSA (Rivest-Shamir-Adleman) - one of the earliest and most widely used.
- ECC (Elliptic Curve Cryptography) - modern, offers strong security with smaller key sizes.

Strengths and weaknesses

- Strength: Solves the key distribution problem - no need to secretly share a key in advance.
- Weakness: Computationally slower than symmetric encryption, so it is inefficient for encrypting large volumes of data.

4. Symmetric vs Asymmetric: Side-by-Side Comparison

Feature	Symmetric Encryption	Asymmetric Encryption
Keys used	One shared secret key	Public + Private key pair
Speed	Fast	Slower
Key distribution	Difficult - key must be shared secretly	Easy - public key can be shared openly
Best used for	Encrypting large amounts of data	Securely exchanging a key / verifying identity
Examples	AES, DES, 3DES	RSA, ECC

5. How HTTPS Uses Both

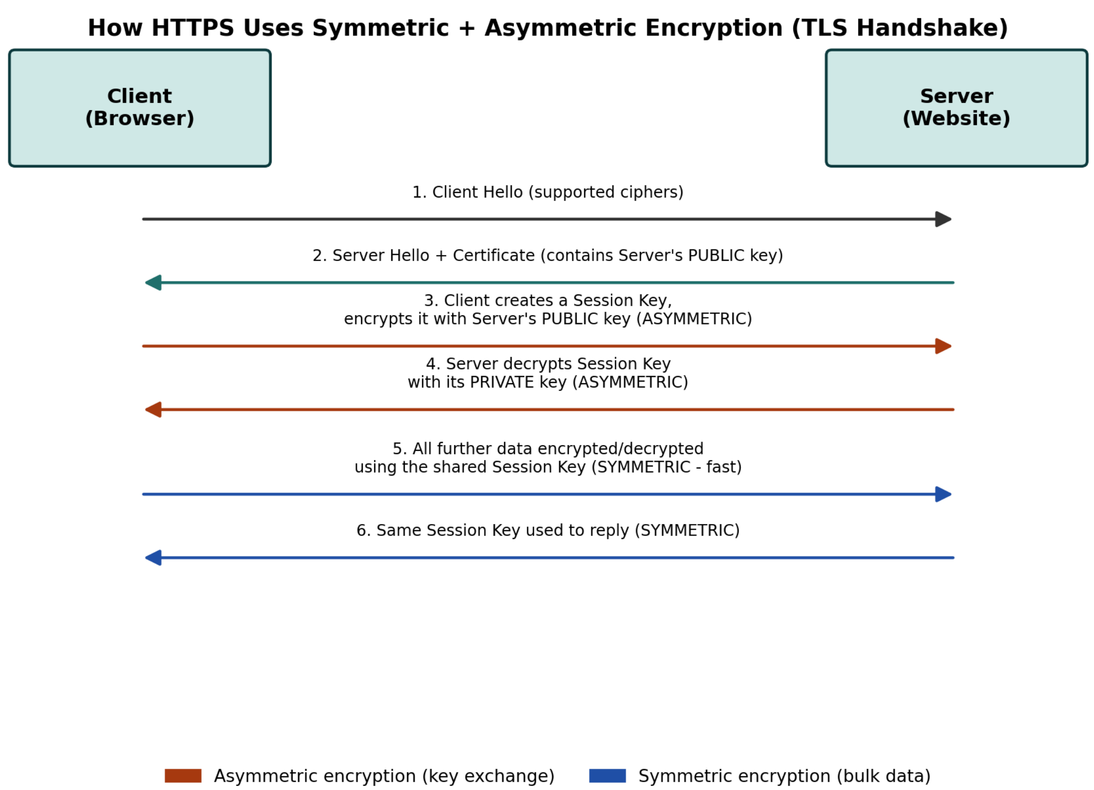
HTTPS (HTTP Secure) protects web traffic using TLS (Transport Layer Security). TLS is designed to combine the strengths of both encryption types: asymmetric encryption is used briefly at the start of a connection to safely agree on a shared secret, and symmetric encryption is then used for the rest of the session because it is much faster.

Step-by-step: The TLS Handshake

- 1. Client Hello: The browser contacts the server and lists the encryption methods it supports.
- 2. Server Hello + Certificate: The server responds with its digital certificate, which contains its public key.
- 3. Session key creation (Asymmetric step): The browser generates a random session key and encrypts it using the server's public key, so only the server can read it.
- 4. Private key decryption (Asymmetric step): The server uses its private key to decrypt and recover the session key.
- 5. Secure data transfer (Symmetric step): Both sides now share the same session key and use fast symmetric encryption (typically AES) to encrypt all further data, such as page content, form submissions, and cookies.

This hybrid approach means HTTPS gets the best of both worlds: the secure key exchange of asymmetric encryption without needing to share secrets in advance, and the speed of symmetric encryption for the actual browsing session.

Diagram: Symmetric + Asymmetric Encryption in HTTPS



6. Conclusion

Symmetric and asymmetric encryption are not competitors but complements. Symmetric encryption offers speed for bulk data, while asymmetric encryption offers a secure way to exchange keys without prior contact. HTTPS - the protocol securing nearly all modern web traffic - is a practical example of this hybrid model in action, using asymmetric encryption briefly to establish trust and a shared key, then switching to symmetric encryption for the remainder of the session.